A Supervised Framework for Recognition of Liquid Rocket Engine Health State Under Steady-State Process Without Fault Samples

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***Abstract*— Data-driven intelligent models have received more and more attention in liquid rocket engine (LRE) state recognition. However, due to practical limitations, the fault data are rare for model training, which results in poor testing results. To better recognize the state of LRE, we proposed a supervised recognition framework without fault samples. First, we proposed a negative sample generation method. Then, fusion recurrent convolutional neural network (FRCNN) is constructed for data fusion and feature extraction. Finally, we evaluated the recognition results with several quantization indicators. The proposed method is verified with the real monitoring data, which is obtained from the static firing tests of a certain type of LRE. Results show that the FRCNN could recognize the state of LRE with accuracy over 93% under the confidence of 96%, which shows the effectiveness of the proposed framework. And the comparison with the literature method shows the superiority of the proposed framework. In addition, it could also perform real-time monitoring for LRE, and recognize the abnormal at the initial stage of the steady state, which has practical potential in monitoring task. At last, we further discuss and analyze the generated negative sample and the proposed framework.**

***Index Terms*— Deep learning, health state, liquid rocket engine (LRE), multi-source data, negative sample.**

I. Introduction

ROCKET engine is the flight power core of carrier rocket, which not only has a very complicated structure, but also has extremely harsh working conditions, including high temperature, low temperature, and high-frequency (HF) oscillation [1]. Under such conditions, any small anomaly can quickly develop into a very destructive failure, resulting in a launch failure and huge losses. Therefore, the recognition of engine health status can, on the one hand, timely find potential hazards to reduce losses and risks and, on the other hand, provide feedback for the optimization and improvement of engine [2].

At present, the commonly used detection methods in engineering applications mainly include the red line method and the expert system method [3]. However, with the application of new technologies, the complexity and quantity of monitoring data have greatly increased. These traditional methods show limitations, such as large detection errors, sharply increased maintenance costs, and lag in detection. In addition, the over-reliance on some important parameters and the neglect of feature extraction on multi-source data also make the traditional methods perform poorly [4].

Recently, based on the accumulation of massive data, deep learning has rapidly developed in engineering applications [5], [6]. Long short term memory (LSTM) network and convolutional neural network (CNN) are applied to the state monitoring and remaining useful life prediction of aero-engine. In addition, transfer learning method has been used for rocket engine anomaly detection [7]. The deep learning method is very suitable for processing multi-source data of the engine and has excellent feature extraction ability [8]. However, these model training generally requires a large amount of normal data and all kinds of abnormal data, but the collected abnormal data are difficult to cover all possible failures, which is difficult to meet the needs of model training. Nevertheless, in actual operation, the possible failures of the engine are complex and diverse. In the absence of fault data, the recognition of engine health status is a challenging problem, which has good potential in practical monitoring tasks.

Fault simulation of liquid rocket engine (LRE) based on physical model is a model-driven method to obtain fault data [9]. For LRE, the model-driven method has high construction complexity and poor flexibility, which makes it difficult to meet actual needs, while for data-driven methods, computational efficiency and real-time performance will be higher. One-class support vector machine (OSVM) is a data-driven method for processing one-class classification problem. OSVM calculates a hypersphere to determine the range of health state and has been used in LRE state recognition [10]. Inductive monitoring system (IMS) [11] is a popular data-driven method in engineering applications. IMS calculates whether the test sample meets the threshold criteria to determine the LRE health state. These data-driven methods can be trained based only on normal data and have the ability

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to recognize the health state of LRE. However, these methods have limited feature extraction capabilities for multi-source data, and the accuracy of the determined classification hyperplane or classification threshold criteria also needs to be improved.

In order to extract better features from multi-source monitoring data and obtain a better classification boundary, we proposed a supervised intelligent framework for recognition of LRE health state without fault samples in this article. First, based on the mechanism of equipment degradation, we propose a data-driven negative sample generation method for model training. Negative sample is designed to help the data-driven model learn a more accurate classification boundary. Then, we propose a fusion recurrent CNN (FRCNN) to learn the normal data (positive samples) and generated negative samples in a supervised way. FRCNN could achieve adaptive fusion of multi-source data and recognize the health state of LRE. In addition, we put forward quantitative evaluation indicators, including recognition confidence and fault degree, to further demonstrate the recognition results. Finally, the proposed framework is verified with the static firing test data of a certain type of LRE. Results show that the running state and fault degree of engine can be identified with a high recognition confidence. The contributions of this article are summarized as follows.

1) Based on the mechanism of equipment degradation, a data-driven negative sample generation method is proposed, which could provide negative samples for supervised model training.
2) FRCNN, trained in a supervised way, is proposed to achieve fusion of multi-source data, feature extraction, and recognition for health state of LRE.
3) The proposed framework is verified with the static firing test data. We also compare it with literature methods to show the superiority of proposed framework.

The rest of this article is organized as follows. Section II focuses on the related works. In Section III, the proposed data generation method and the entire framework are introduced. Section IV shows the case study. Then, we further discuss the data generation method and proposed framework in Section V. Finally, the conclusion is drawn in Section VI.

II. Related Work

With the development of rocket engine technology, the amount and complexity of monitoring data continue to increase. At the same time, the research on LRE state recognition has also been continuously developed to provide decision support for LRE health management [12].

Red line method and the expert system are classic state monitoring methods [13]. However, with the application of new technologies such as high-thrust rockets and recoverable rockets, the informatization and complexity of rockets have been greatly improved, and many limitations of these methods have been revealed, such as large detection errors, sharply increased rule maintenance costs, and delayed detection time.

In recent years, based on the accumulated massive data, new technologies such as deep learning have been applied in many engineering applications field, such as machine vision and natural language processing [12]. In the field of system health monitoring, most research has focused on the fault diagnosis and performance evaluation of certain key components (such as bearings and gearboxes) [13]. However, for the health monitoring of complex systems, especially in the aerospace industry, there is little research with deep learning methods [14]. Aiswarya *et al.* [15] manually extracted the time and frequency domain features of the LRE running signal, and used support vector machines to classify these features, and achieved 100% classification accuracy. Xu *et al.* [4] proposed a back propagation (BP) neural network optimized by quantum genetic algorithm for fault diagnosis of LRE. Yuyang *et al.* [16] proposed a fault detection method with particle swarm optimization and least square support vector machine to improve the performance for LRE. Wang *et al.* [17] proposed a deep separable CNN to predict the remaining useful life of aero-engine with the monitoring data from different sensors. Miao *et al.* [18] proposed the dual-task deep LSTM networks to unify the task of degradation evaluation and remaining useful life prediction of aero-engine.

IMS [11], proposed in 2004, is designed to monitor the operating state of LRE without providing specific diagnostic results. IMS is based on a clustering algorithm, which only needs to use normal data for training. For further analysis and interpretation of IMS, Martin *et al.* [19] simulated several kinds of fault data and used it to make a preliminary verification of the fault monitoring capability of the system. Then, they used IMS to test the actual operation data of the space shuttle Ares I-X, and discovered the ability of IMS to monitor unknown anomalies and potential failures. Based on its highly reliable and stable anomaly monitoring capabilities, IMS has been successfully applied in many aerospace fields [20], such as International Space Station Mission Control, TacSat-3, Ares I-X, and Launch System Ground Support Equipment. In summary, the practicality and reliability of IMS mainly benefit from the specific strategy of its algorithm. IMS makes the model to learn the boundary of health state to realize the identification of abnormal state, which is a good inspiration for the design of anomaly monitoring algorithms.

IMS is a one-class data-driven anomaly detection method, which relies on only one class for training—normal data. It is appropriate when little or no fault data are available for training, which is usually the case in space applications. Zhu *et al.* [10] proposed a fault detection framework for the steady-state process of LRE based on a convolutional auto-encoder and an OSVM. In addition, its results show that by increasing sample size and introducing domain knowledge, the fault detection method could be further improved. When more numbers and categories of fault data are available, using multiclass data-driven methods, such as supervised classification method, could obtain specific diagnosis results rather than abnormal monitoring. Schwabacher *et al.* [21] proposed a supervised learning method to diagnose fault in the J-2X rocket engine. By training with a large number of data, it can obtain higher accuracy and interpretability. However, if no training data are available, especially for the new system, functional model-based approach could get better performance [22]. It could encode the expert knowledge about the operation of a system into a model, and provide a diagnosis result with better detailed explanation. It is often beneficial to use multiple monitoring strategy for specific application [23].

III. Proposed Method

In this section, we would detail the proposed framework, including the two methods for negative sample generation as well as data fusion and feature extraction based on FRCNN. With the framework, the state of LRE could be recognized without fault samples.

In the process of data analysis and research, we found that the abnormal data obtained in experiment are limited, and it is impossible to obtain all the abnormalities that might occur in the actual running. Therefore, it is difficult to recognize other abnormalities in actual running with the model trained with the abnormal data obtained from experiment. Based on the principles of anomaly detection methods (IMS, OSVM, etc.), we proposed a supervised framework to recognize the possible abnormalities for LRE in the actual running process without fault sample. With the help of generated negative samples, we could train the model to learn a better boundary for the normal data in a supervised way.

A. Negative Sample Generation

In this section, we propose a data-driven negative sample generation method, which could provide the data-driven model training with an infinitely large set of negative samples. It is designed based on the failure mechanism and degradation law. We generate samples by superposition, which not only retains the components of the normal sample, but also adds the abnormal feature. In this way, the generated samples could maintain a relatively close distance from the normal samples. Negative sample generation methods include vibration sharpening and resampling superposition.

1) Vibration Sharpening: For multi-source vibration data, different channels have a unified time series. The sample of multi-source signal is formulated as follows:

$$M = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_S \end{bmatrix} = \begin{bmatrix} x_{1,1}, x_{1,2}, & \cdots & x_{1,N} \\ x_{2,1}, x_{2,2}, & \cdots & x_{2,N} \\ \vdots & \ddots & \vdots \\ x_{S,1}, x_{S,2}, & \cdots & x_{S,N} \end{bmatrix} \tag{1}$$

where M represents a sample, X_i is sample from a certain channel, $x_{i,j}$ is collected data from one channel, S is the number of channels, and N is the length of sample.

The change of engine state could be reflected in time-domain signals, such as peaks, vibrational energy, and local rise and fall gradients [3]. Therefore, we propose a negative sample generation method, vibration sharpening, from the perspective of time domain, which is formulated as follows:

$$X_A = \eta_A\left[\lambda_A X_G + (1-\lambda_A)\frac{\operatorname{sign}(X_G)|X_G|^\gamma}{\max(|X_G|^\gamma)}\right] \tag{2}$$

where $X_G = [x_1^G, x_2^G, \ldots, x_{N-1}^G, x_N^G]$ is the channel data of a normal sample, $X_A = [x_1^A, x_2^A, \ldots, x_{N-1}^A, x_N^A]$ is the generated data, η_A is the scaling factor and $\eta_A \in (0.5, 1.5)$, λ_A is the mixing factor and $\lambda_A \in [0, 1)$, γ is the sharpening factor and $\gamma \in (1, 3]$, sign$(\cdot)$ is a function that takes the sign of the number in the sequence, and max$(\cdot)$ is used to take the maximum value in the sequence.

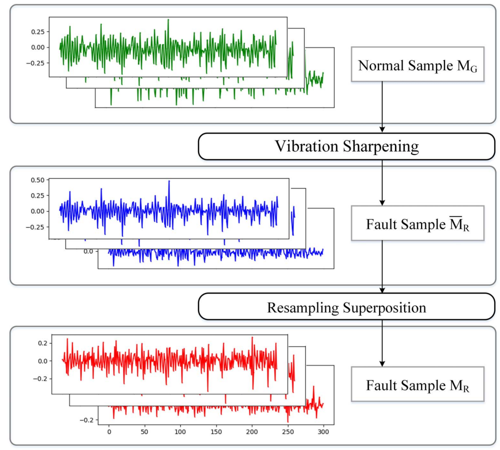

Fig. 1. Flowchart of negative sample generation.

Through vibration sharpening, the local rise and fall gradients of time-domain signal could be changed to simulate the vibration amplitude change caused by engine abnormal. By selecting different factor parameters, different degrees and modes of abnormal can be generated. For example, η_A can adjust the amplitude of generated signal, because in the rocket engine, the amplitude change caused by the abnormal is diverse.

2) Resampling Superposition: Generally speaking, when the state of engine becomes abnormal, two characteristics will appear on the corresponding spectrum [17]. One is the increase in spectrum amplitude, including an increase at multiple frequencies or an increase at a single frequency (SF), which could be partially achieved by vibration sharpening. Another feature is the change of the spectrum type, that is, the appearance of a new single component in the spectrum, or the appearance of a new peak group in the frequency band, which would be achieved by resampling superposition.

Resampling superposition is expressed as follows:

$$X_F = \eta_F[\lambda_F X_G + (1-\lambda_F)\text{resample}(X_G, f_t, f_s)] \tag{3}$$

where $X_F = [x_1^F, x_2^F, \ldots, x_{N-1}^F, x_N^F]$ is the generated data, η_F is the scaling factor and $\eta_F \in (0.5, 1.5)$, λ_F is the mixing factor and $\lambda_F \in [0, 1)$, resample$(\cdot)$ is the resampling function in MATLAB, f_t is the target sampling frequency, and f_s is the original frequency.

With the resampling superposition, new frequency features components can be added to the generated samples.

3) Negative Sample Generation Method: During running process of LRE, failures may occur in a certain component or the entire engine. We need to provide sufficient and diverse negative samples, which could form a close boundary with normal samples to help the model learn a more accurate range. For this purpose, we propose the negative sample generation method, as shown in Fig. 1, based on vibration sharpening and resampling superposition, which includes the following steps.

1) A normal sample M_G is randomly chosen. Then, we randomly choose an integer s_A from the interval $[1, S]$ and randomly select s_A channels from M_G. Next, s_A channels are processed with vibration sharpening to obtain sample $\bar{M}_R$.
2) We randomly choose an integer s_F from the interval $[1, S]$ and randomly select s_F channels from the above sample $\bar{M}_R$. Next, the s_F channels would be processed with resampling superposition method to obtain a negative sample M_R.

Finally, we could get generated negative samples with different abnormal channels, different abnormal degrees, different abnormal types, and so on. The generated negative samples have the following characteristics.

1) The negative samples generation methods are based on the failure mechanism, and the process is interpretable.
2) By adjusting the hyper-parameters, we could provide an infinitely large set of generated samples, which could form a close boundary around the normal samples.
3) Most generated negative samples may not correspond to the actual failure type. The proposed method is not to generate very real fault samples, but to provide sufficient negative samples for model training.

B. Data Fusion and Feature Extraction With RCNN

With the above method, we could provide an infinitely large set of negative samples for model training. For multi-source monitoring data, valid anomaly information may be included in any channels. Through learning the multi-source data, we might find the interaction of multi-source data that are difficult to notice from a single channel data [10]. Therefore, we proposed the multi-source data fusion method to retain and mine the features of different channels.

In order to achieve data fusion and feature extraction, we proposed an FRCNN, which is trained in a supervised way based on the generated negative samples. Besides, the multi-source data fusion and feature extraction could be achieved in one model. Therefore, FRCNN could achieve adaptive fusion by giving higher weight to important channels and realize the health state recognition of LRE. The structure and parameter settings of network are shown in Table I.

First, the LSTM layer is used to realize the adaptive fusion for multi-source data and extract the state information hidden in time sequence. Then, the convolution (CL) layer and max-pooling (MP) layer are added to adaptively extract features. Besides, we add the batch normalization (BN) layer to improve the feature extraction ability. In this way, the intrinsic features of the engine running state are extracted. Finally, with the dense layer, FRCNN could realize the non-linear mapping of the intrinsic features and the running state and achieve state recognition of the sample.

C. Procedure of the Proposed Framework

Based on the negative sample generation methods and proposed FRCNN, the supervised framework is proposed to recognize the LRE health state without fault samples. The procedure of proposed framework, shown in Fig. 2, is introduced as follows.

1) Negative Sample Generation: First, we need to pre-process the multi-source data and obtain normal samples. Then, with the proposed negative sample generation method, we could provide an infinitely large set of negative samples for model training.

2) Recognition for Health State of Sample: The FRCNN is trained with the normal samples and generated negative samples. The trained model is used to recognize the state of testing samples. Finally, the recognition results are evaluated with several quantization indicators, which would be introduced in Section IV.

TABLE I
DETAIL OF NETWORK STRUCTURE

No.	Layer	Kernel Number	Kernel Size	Stride	Output Shape Width×Depth	Padding
1	Input				2048×19	
2	LSTM_1				2048×64	
	LSTM_2				2048×1	
3	Convolution_3	16	32	4	512×16	same
	BN + ReLU					
	MaxPooling_3	16	2	2	256×16	same
4	Convolution_4	32	4	1	256×32	same
	BN + ReLU					
	MaxPooling_4	32	2	2	128×32	same
5	Convolution_5	64	4	1	128×64	same
	BN + ReLU					
	MaxPooling_5	64	2	2	64×64	same
6	Fully-Connected				64×1	
	Fully-Connected				2×1	

IV. CASE STUDY

This section presents a case study with the multi-source monitoring data from static firing tests of a certain type of LRE, which is used to verify the effectiveness of the proposed framework. The case study includes three parts: dataset preparation, model configuration and training, and verification of the framework.

A. Dataset Preparation

1) Dataset Description: The dataset for experiment is collected from the static firing test process of a certain type of LRE. Fig. 3 shows the main structures and working process of the LRE used in the static firing test process.

The dataset consists of 18 groups of data, including 13 groups of normal data and five groups of abnormal data, where 19 common channels are selected, as shown in Table II. Each group is the data obtained by a complete static firing test experiment of one engine (newly assembled one or previously operated one), and different time lengths can be set according to the experimental design.

Among the dataset, the fault group data are mainly collected from the early stage of engine design. With the optimization of the design, in the later experiments, failures are relatively

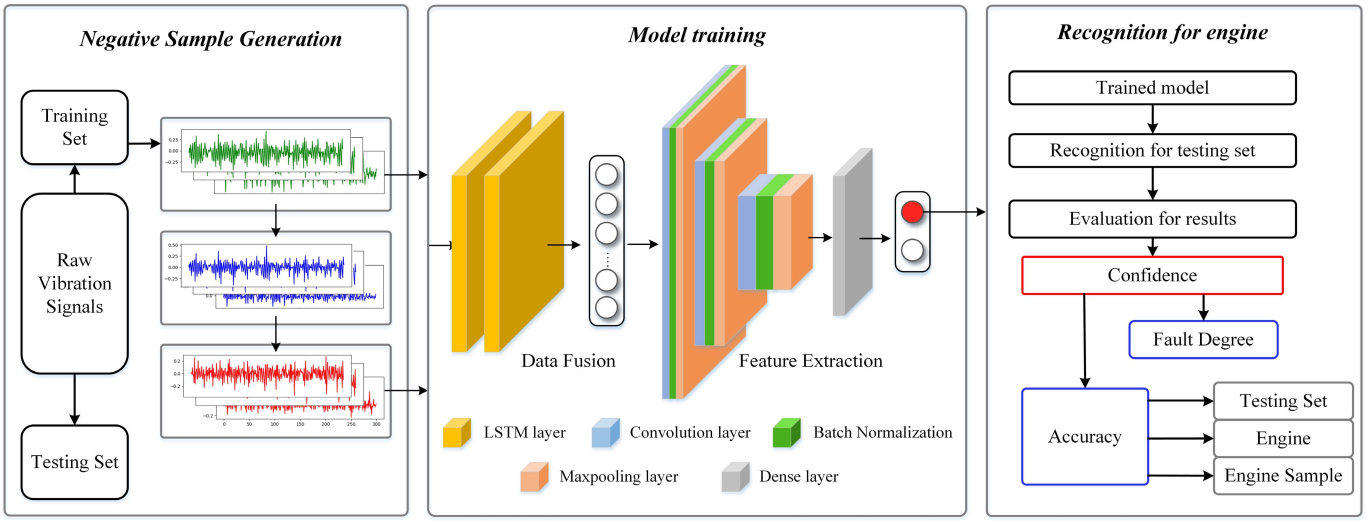

Fig. 2. Flowchart of the proposed framework. It consists of three stages: negative sample generation, model training, and recognition of engine.

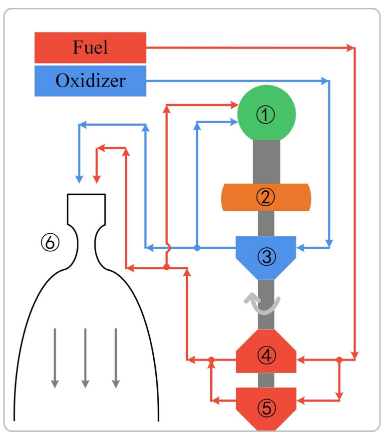

Fig. 3. Main structures and working process of the LRE. (① Gas generator, ② main turbine, ③ oxygen main pump, ④ fuel primary pump, ⑤ fuel secondary pump, and ⑥ thrust chamber.).

TABLE II
SELECTED PARAMETERS

No.	Symbol	Parameter	Unit
1	a_{c1}	Axial vibration of thrust chamber	m/s^2
2	a_{c2}	Radial vibration of thrust chamber	m/s^2
3	a_{c3}	Tangential vibration of thrust chamber	m/s^2
4	a_{gg1}	Axial vibration of gas generator	m/s^2
5	a_{gg2}	Radial vibration of gas generator	m/s^2
6	a_{gg3}	Tangential vibration of gas generator	m/s^2
7	a_{po1}	Axial vibration of oxygen pump	m/s^2
8	a_{po2}	Radial vibration of oxygen pump	m/s^2
9	a_{po3}	Oxygen pump tangential vibration	m/s^2
10	a_{ppo1}	Axial vibration of oxygen pre-pressure pump	m/s^2
11	a_{pf}	Radial vibration of fuel secondary pump	m/s^2
12	p_{io}	Oxidizer inlet pressure	MPa
13	p_{eppo}	Outlet pressure of oxygen pre-pressure pump	MPa
14	p_{epo}	Oxidizer pump outlet pressure	MPa
15	p_{ihog}	Generator pressure before spraying	MPa
16	p_{if}	Fuel inlet pressure	MPa
17	p_{epf1}	Fuel primary pump outlet pressure	MPa
18	p_{ihfg}	Generator pressure before fuel injection	MPa
19	p_{ihfc}	Combustion chamber pressure before fuel injection	MPa

fewer. The specific fault type of the abnormal data is unknown. For the data labels, they are marked according to expert experience, real-time monitoring, and post-mortem maintenance. Therefore, the reliability of label is high.

2) Data Preprocessing: There are three main stages in engine operation: starting stage, steady-state stage, and shutdown stage. In this article, we focus on analyzing the data at steady-state stage. It has a longer running time and a relatively stable running state, which contains rich information about LRE state.

First, we intercept the steady-state stage data from each group and clean up the dirty point in signal. Then, data of identical channels from all groups would be jointly normalized, which is expressed as follows:

$$\tilde{T}_j = \frac{T_j}{\max\left(\left|T_j\right|\right)} \tag{4}$$

where T_j is the jth channel for all groups of monitoring data, and $\tilde{T}_j$ is the corresponding result after normalization, $|\cdot|$ is used to obtain the absolute value of each point in the sequence, and $\max(\cdot)$ is the function that takes the maximum value of the sequence.

In this case, 100 samples are collected from each group, and the samples need to cover the whole sequence evenly. Each sample covers 2048 consecutive points. The dataset is divided and distributed according to the group. Eight groups of normal data are selected for training set, and the other five groups of normal data and fault data are used for testing.

3) Negative Sample Generation: For training set, the negative samples are generated during each epoch of the training process. There are 800 normal samples, but the generated samples are random. In this way, we could provide the model

TABLE III
HYPER-PARAMETERS SELECTION

Hyper-parameter	Selectable values	Selection results
Learning rate	10^{-4}, 10^{-3}, $5*10^{-3}$, 10^{-2}	$5*10^{-3}$
Batch size	8, 16, 32, 64	16
Optimizer	Adam [25], SGD [26]	Adam

with an infinitely large set of generated samples (randomly selected normal samples, randomly selected channels, and randomly selected abnormal factors). The number of positive samples and negative samples input to the model in each epoch is equal, which could help the model get better recognition results. In addition, we also generate negative samples based on the normal samples of testing set, which is also used for testing.

Finally, in training set, we get 800 normal samples and an infinitely large set of generated negative samples, in test set, we obtain 500 normal samples, 500 generated samples, and 500 fault samples.

B. Model Configuration and Training

Experiments are run on a personal computer with Intel Core i5-9400F (2.90 GHz) CPU (16-GB memory), Geforce GTX 1050 Ti GPU (4-GB memory), and Microsoft Windows 10 ultimate operation system. Programming language is Python 3.7 with scientific computing library "Pytorch" (1.2.0) [24].

For hyper-parameter selection, Table I shows the selectable values. Based on different combinations of the hyper-parameter, we use FRCNN to learn the training set. By comparing the training effect, we selected a combination of hyper-parameters that performed better, which is shown in Table III.

After training and comparison, we chose a larger learning rate to enable the model to update the weights faster in the early stages of training and avoid the model from falling into the local optimum. Besides, a smaller batch size is selected, mainly to avoid generating too many negative samples with large differences in a batch, which could improve the feature learning effect of model. For the optimizer, we finally choose the Adam, which performs better during the model training on this case with higher convergence speed and stability.

After the model configuration, we would train the model based on the normal samples and generated negative samples. The FRCNN is trained for about 200 epochs until the loss converges.

C. Verification of the Framework

1) Evaluation Method: In this article, we proposed three indicators to quantify the recognition results, including confidence, fault degree, and recognition accuracy.

During the training, the normal data are labeled as 0 and the fault data as 1. The network output before Softmax layer, value of the first node, is used as the recognition result. For a sample M, the recognition result is formulated as

$$y = \mathrm{FRCNN}(M) \tag{5}$$

where FRCNN represents the network and y is the recognition result of sample M.

The classification threshold between normal and fault class is denoted by $\tilde{y}$, which is calculated from the following formula:

$$\begin{cases} \mathrm{Conf} = \dfrac{\mathrm{Num}_{k=1}^{Q^{\mathrm{Train}}}\left(y_k^{\mathrm{Train}} \le \tilde{y}\right)}{Q^{\mathrm{Train}}} \times 100\% \\ \tilde{y} \in \left\{y_k^{\mathrm{Train}} | k \in [1, Q^{\mathrm{Train}}]\right\} \end{cases} \tag{6}$$

where Conf is the confidence for the recognition results, $\mathrm{Num}(\cdot)$ is the function of counting, Q^{Train} represents the number of normal samples in training set, and y^{Train} is the recognition results of the normal sample in training set.

With a certain Conf, if the recognition result y of a sample is larger than $\tilde{y}$, it would be recognized as fault sample. In addition, the mean value of the recognition results of the samples from the same group is denoted by $\bar{y}^G$, which is calculated as follows:

$$\bar{y}^G = \frac{1}{Q^G}\sum_{k=1}^{Q^G} y_k \tag{7}$$

where Q^G is the number of samples from a certain group. If $\bar{y}^G$ is larger than $\tilde{y}$ under a certain Conf, the corresponding engine is classified as a fault one.

After training, the recognition results of normal data are close to 0, most of which are less than 0.1. While the output of generated negative samples is about 0.35, the actual fault sample was between 0.15 and 0.35. According to the values of the output results of the model, fault degree Deg is used to quantitatively describe the fault severity of a certain fault engine, which is formulated as follows:

$$\mathrm{Deg} = \mathrm{ceil}(10 \times (\bar{y} - \tilde{y})) \tag{8}$$

where $\mathrm{ceil}(\cdot)$ is the function to round up.

For recognition accuracy, it can be divided into three results sample from different perspectives: recognition accuracy of testing set, state recognition accuracy of engine, and sample recognition accuracy of each engine. First, we need to count the recognition accuracy of all testing samples and record the accuracy as sample recognition accuracy of testing set. Second, we would classify the running state of each engine and calculate the state recognition accuracy of all engine. Third, the sample recognition accuracy of each engine is calculated.

Through the above quantitative evaluation indicators, we can make multi-dimensional, more comprehensive recognition results for the testing set.

2) Recognition Results: After training, the trained model is used to recognize the state of each sample in testing set. Finally, the results are shown in Table IV. We set the confidence level to over 90% to ensure the reliability of the results. The analysis of the results is summarized as follows.

1) Under the confidence of 93%–99%, the trained FRCNN could accurately recognize the state of engine, including the normal engine and fault engine.
2) With the increase of confidence, the fault degree would decrease, especially when 96% increases to 99%. The sample recognition accuracy for normal engine keeps

TABLE IV
EVALUATION FOR RECOGNITION RESULTS

State of engine	No.	Recognition of state			Fault degree			Sample recognition accuracy of engine		
Confidence		93%	96%	99%	93%	96%	99%	93%	96%	99%
Normal engine	1	N	N	N	0	0	0	90%	95%	99%
	2	N	N	N	0	0	0	83%	89%	96%
	3	N	N	N	0	0	0	74%	81%	93%
	4	N	N	N	0	0	0	98%	98%	98%
	5	N	N	N	0	0	0	100%	100%	100%
Fault engine	6	F	F	F	2	2	1	98.8%	98.5%	72.4%
	7	F	F	F	2	2	1	95.6%	90.4%	81%
	8	F	F	F	2	2	1	100%	99.7%	96.7%
	9	F	F	F	3	3	2	95.7%	94.0%	93.2%
	10	F	F	F	2	2	1	89.2%	86.3%	70.4%
Total recognition accuracy		100%	100%	100%		/		92.4%	93.2%	90%
Generated negative samples	(1)	F	F	F	3	3	2	99%	98%	98%
	(2)	F	F	F	3	3	2	99%	99%	99%
	(3)	F	F	F	3	3	2	98%	98%	98%
	(4)	F	F	F	3	3	2	100%	100%	100%
	(5)	F	F	F	3	3	2	100%	100%	100%

TABLE V
RESULTS OF LITERATURE METHODS

Method	IMS	OSVM	Proposed
Accuracy	82.5%	88.1%	93.2%

increases, while it decreases for fault engine, especially for engines #6 and #10.

3) The best sample recognition accuracy of testing set is 93.2% when confidence is set to 96%. With this confidence, the recognition accuracy of normal engine and faulty engine is balanced.

During the testing process, we also evaluated the recognition results for generated samples to test the representation capability of the generated sample. The negative samples were generated based on normal group sample of the test set. As shown in Table IV, the model could identify the generated samples as a group with a high degree of abnormality, which also indicates that the trained model has satisfactory anomaly recognition capability, and the generated samples have good anomaly representation capability.

We also compared with the popular LRE anomaly literature detection method using the same training set and testing set, and the results are shown in Table V. We refer to the parameter selection in the literature to configure and optimize the model. For IMS [27], the clustering interval is set to [0.5, 0.6]. For OSVM [10], nu is set to 1×10^7, and gamma is 0.1.

Results show the superiority of the proposed supervised framework, which could learn a better boundary for normal class. Compared with IMS, deep learning methods have better feature extraction capabilities. For OSVM, the proposed method combines data fusion and state recognition in one model for training, which can better adaptively extract features in a supervised training method.

3) Real-Time Monitoring With the Framework: In the above experiments, the effectiveness of proposed framework is verified. It is worth noting that the time span of each sample is short, only 0.08 s. It can be considered that the sample reflects the state of the engine at a certain moment. And the structure of the model is light, and experiments show that the calculation time for a single sample is less than 0.1 s. In addition, by learning transient samples, the model can make a timely response to the state of the engine and deal with the problem of missing and incomplete data. Therefore, the proposed framework has the potential for real-time monitoring.

For actual monitoring, we collect m samples from the multi-source signal per second and perform real-time identification. In order to improve the stability of monitoring, we set an alarm threshold ε, when the number of consecutive faults exceeds ε, and a fault alarm is issued. Generally, ideal real-time monitoring results have two requirements. For normal engine, the model can avoid misjudgment and even detect potential anomalies. For fault engine, the model can detect and forecast the anomaly as early as possible.

In order to verify the real-time monitoring capability of the proposed framework, we designed the following experiment. Take engines #4 and #6 in Table IV as monitoring objects, we collected one sample per second of the signal. And the alarm threshold is set to two. The collected sample includes the starting stage, steady-state stage, and shutdown stage. And the recognition results during the whole monitoring process are shown in Fig. 4. In order to observe the monitoring results intuitively, we have marked the value of $\tilde{y}$ under different confidence in the figure.

The FRCNN is trained with the data of steady-state stage, so the results from starting stage and shutdown stage are recognized as abnormal, while for the monitoring of steady-state stage, the recognition results are satisfactory, which is summarized as follows.

1) For normal engine, the misjudgment for FRCNN is less than 2%, and there is no continuous misjudgment exceeding the threshold two. Results show that FRCNN has learned features with higher robustness in normal engines.

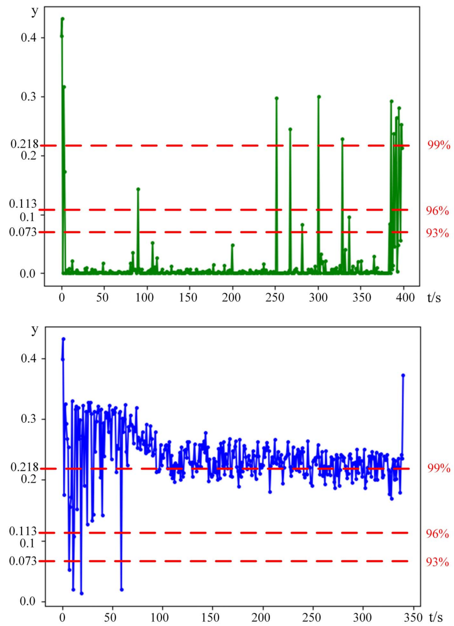

Fig. 4. Real-time recognition results. Top: normal engine #4. Bottom: fault engine #6.

2) For engine #6, the abnormal is identified after 300 s in actual monitoring. But FRCNN could identify the abnormal when the engine initially enters steady state, which can recognize the fault before it develops to a serious stage.

V. Discussion

In this section, we would further discuss and analyze the effectiveness of the generated negative sample and then discuss the advantages and disadvantages of the proposed framework in detail.

A. Discussion on Generated Negative Sample

Negative sample generation method is the foundation of proposed framework. The generated samples help the FRCNN learn a better boundary for normal state and achieve the recognition of state for LRE. In this part, we would further study the role of generated sample.

The generated methods include a time- and frequency-domain methods. As shown in Fig. 5, we compared the generated samples with the fault samples from time-domain waveforms and spectrum. Results show that the generated samples are obviously different from the original signal. In the time domain, the statistical parameters of the generated sample are different from the original one. In the frequency domain, the generated sample not only retains the frequency characteristics of the original sample, but also has new frequency characteristics.

In multi-classification tasks, generative adversarial networks (GAN)-based data generation methods can achieve good results [28]. But it requires that the generated classes have the same distribution as the real data. And it is difficult for us to generate effective data through GAN in this situation. For

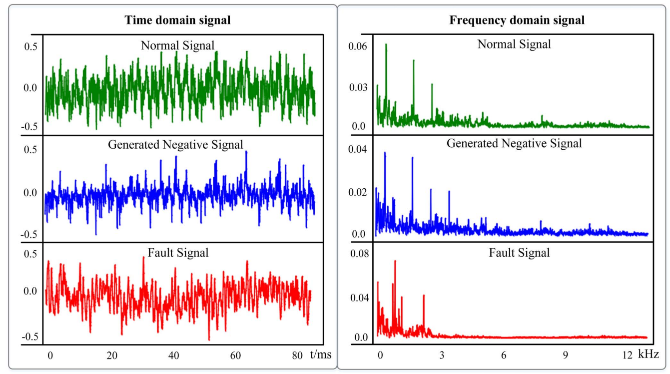

Fig. 5. Waveform of the time-domain signal of channel 17. Vertical axis is the signal amplitude after normalization, which has no specific physical meaning.

monitoring abnormal problems, such as OSVM [29], the most important thing is to learn the boundary of normal class. With the proposed framework, we trained the deep learning model in a supervised way to better extract features from samples and get better recognition results.

Most generated negative samples may not correspond to the actual failure type. However, we would provide the model with an infinitely large set of negative samples, which could form a close boundary around the normal samples. The generated samples maintain a closer distance to the normal sample (as much as possible less than or equal to the distance between the actual abnormal data and the normal data) to help the model learn better border.

B. Advantages and Disadvantages

The advantages of the proposed framework are summarized as follows.

1) We proposed a data-driven sample generation method, which does not require a labor-intensive modeling process and does not depend on the physical modeling and mechanism of the engine.
2) The deep learning model is trained in a supervised way based on the generated negative samples; besides, we combine multi-source data fusion and feature extraction into one model for training instead of training separately [10]. Therefore, the proposed framework could give higher weight to important channels to achieve adaptive fusion and get better recognition results.
3) The proposed framework mainly provides monitoring rather than specific diagnosis. It has the ability to detect anomalies that are previously unknown and have not been simulated or explained before. It is sensitive to abnormalities and can monitor early faults and abnormalities. Even potential weak faults in normal data may be detected.

However, the proposed framework has some disadvantages that need to be studied. First, the generation of fault samples is based on data-driven method, and the theoretical basis of this method needs to be improved. We could deeply study the fault features of different channels to generate fault samples that are more similar to real data. Besides, despite the high accuracy of the recognition results, the interpretability of the model itself is poor. By combining with expert system or other monitoring systems, it might provide specific diagnosis results on the basis of monitoring. Last but not least, we need to improve the generalization ability of the model as much as possible, so that it can perform recognition for health state on more types of LRE.

VI. Conclusion

In this article, we proposed a framework for recognition of LRE health state without fault samples. First, we obtain the negative samples with the proposed sample generated method. Then, the proposed FRCNN is trained with the normal sample and generated negative samples. Finally, the trained FRCNN is used to recognize the state of testing samples. Besides, we perform a multi-dimensional quantitative evaluation for the recognition results. With the static firing test monitoring data of LRE, the method is verified, which achieves accuracy over 93% under the confidence of 96%. The comparison results with literature method also show the superiority of the proposed framework. The trained FRCNN could also realize real-time monitoring and recognize system abnormalities in time. With the sample generation, data fusion, and feature extraction, the proposed framework has practical potential in monitoring tasks.

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